Midterm information, practice questions, common mistakes on HW1

PSTAT126 Lab 5

Time and Place

We will have an **in-class midterm exam** in the lecture room from **9:30 AM to 10:45 AM on May 7**.

Please remember to bring a blue book for your answers. The exam is open-book and open-notes, and you may use a laptop or other electronic devices. However, you must not share or discuss your exam with anyone before the end of the exam. There will be no make-up midterm.

Exam Scope

The midterm exam will cover everything we learned in **Handout 1 to Handout 3**. Specific topics include:

1. **Definition and assumptions for linear regression** with one predictor variable, properties of the model, interpretation of parameters.
2. **Least squares and maximum likelihood estimation of parameters**, properties of the estimates.
3. **Confidence interval and hypothesis test for parameters**.
4. **Estimation of the mean response** and prediction of a new observation and confidence intervals.
5. **ANOVA table**.
6. **Coefficient of determination**.
7. **Residuals and fitted responses**, diagnostic plots, outliers.
8. **Transformations**, Box-Cox transformation.
9. **R function for fitting a linear regression model**, understanding R output.

Questions

The exam will consist of **3 parts**:

1. A question about the **simple linear regression model**: definition, assumptions, interpretation, possible application, R function, etc.
2. A set of **true and false questions** .
3. Questions based on **R output** .

Some example True of False questions

1. The intercept of the simple linear regression model represents the expected value of the dependent variable Y when the independent variable X is zero.

2. The least squares regression line minimizes the sum of the absolute differences between the observed and predicted values of Y.

3. The slope of the regression line remains unchanged if all values of the independent variable X are scaled by a constant.

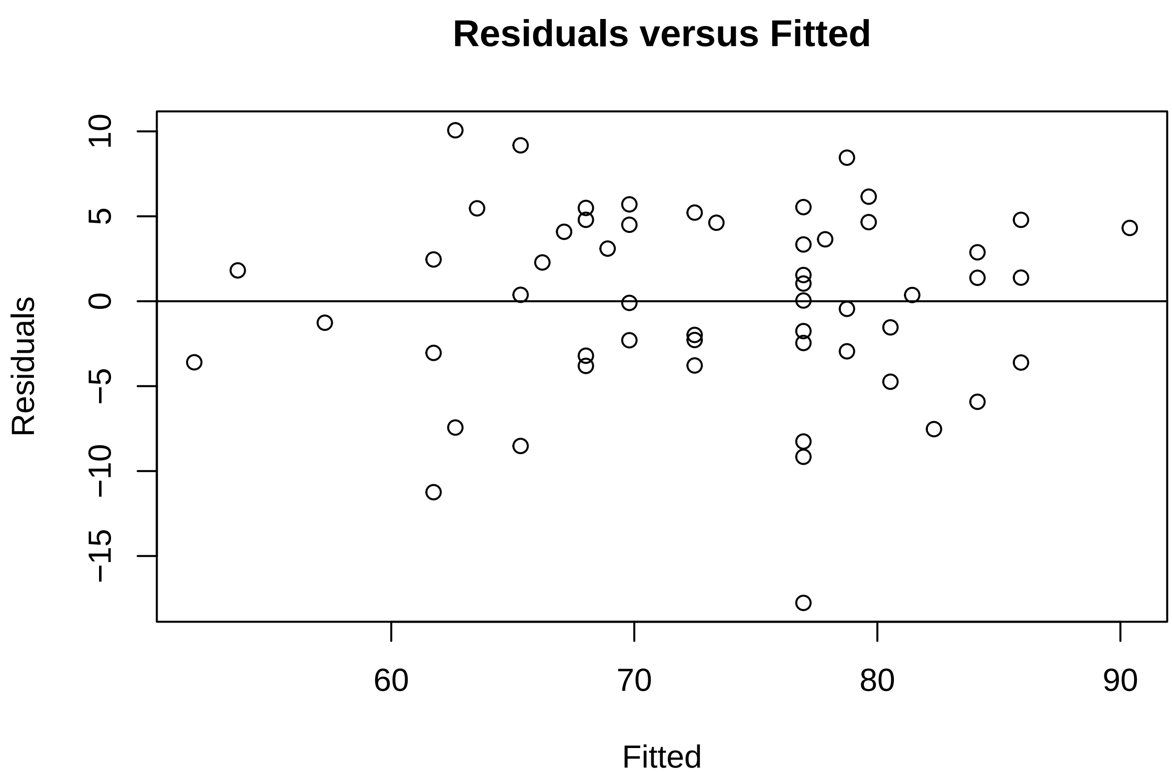
4. The residuals in a simple linear regression model are the differences between the observed values and the predicted values of Y.

Example question based on R output

```r
library(faraway)
x = stat500$midterm
y = stat500$total
fit1 = lm(y ~ x)
summary(fit1)
```

```
##
## Call:
## lm(formula = y ~ x)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17.7587  -3.1245   0.3779   4.4083  10.0633
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   36.678      3.255   11.27 1.09e-15 ***
## x              1.790      0.156   11.48 5.47e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.496 on 53 degrees of freedom
## Multiple R-squared:  0.7131, Adjusted R-squared:  0.7077
## F-statistic: 131.7 on 1 and 53 DF,  p-value: 5.465e-16
```

```r
plot(fitted(fit1), residuals(fit1), xlab = "Fitted", ylab = "Residuals")
abline(h = 0)
title("Residuals versus Fitted")
```



(a) Write down the model being fitted and the fitted regression model.

The fitted regression model is:

where:

(b) State the hypotheses for the t-test for the slope. Interpret the results of the test based on the output. Use significance level $\alpha = 0.5$.

(c) Comment on any patterns in the plot and explain whether the assumption of constant variance is met.

Comments on the Plot:
- The plot should show random scatter with no patterns if assumptions are met.
- If patterns exist (e.g., funnel shape, curvature), this may indicate violations of homoscedasticity(constant variance) or linearity.

Common mistakes in homework 1

Proof of Q5

Please check the homework 1 solution (posted on Canvas) and lecture handout 1, page 35 - 37 (similar).

2c and 3c

To fit a model without an intercept, use the formula:

```
lm(y ~ x - 1)
```

Here, the `- 1` in the formula explicitly removes the intercept term from the model. This fits a regression model of the form:

$$Y = \beta_1 X + \epsilon$$

To fit a model with only an intercept and no predictors, use the formula:

```
lm(y ~ 1)
```

This fits a regression model of the form:

$$Y = \beta_0 + \epsilon$$

.

Additional common mistakes

Not reporting or commenting on results and skipping steps in the solutions.